

## Activity 3.1.1 Illumination Creation

# PLTW

### ILLUMINATION CREATION

---



Lighting the Way



Building Circuits



Unit 3 Problem Introduction



Conclusion

# Lighting the Way

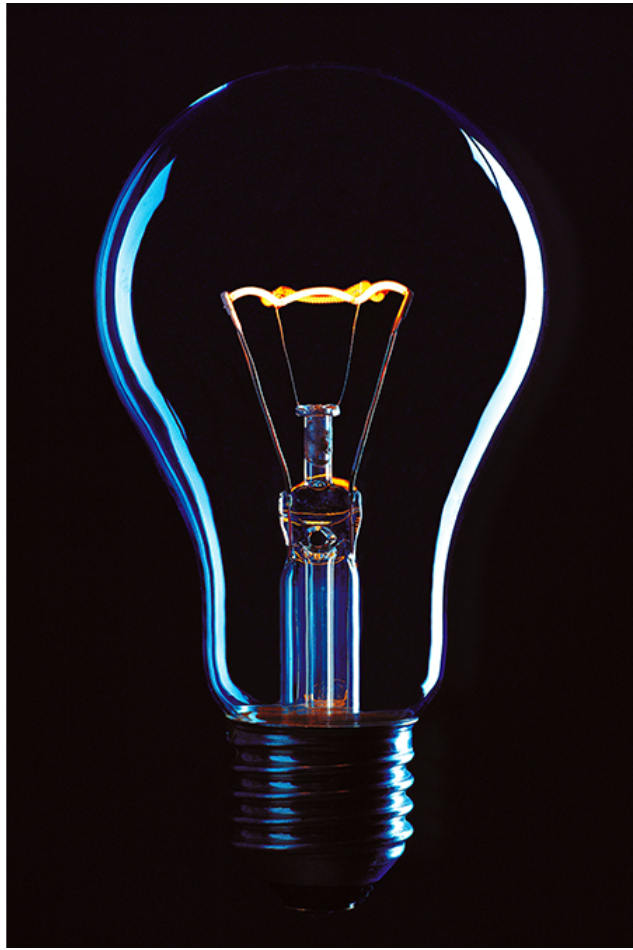
---

| GOALS                                                                                                                                                                                                                                                                                                                      | MATERIALS | RESOURCES |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| <ul style="list-style-type: none"><li>• Explore the relationships between various circuit components.</li><li>• Draw diagrams that show proposed electrical configurations</li><li>• Generate explanations about correct and incorrect electrical configurations.</li><li>• Contribute to the efforts of a team.</li></ul> |           |           |
| GOALS                                                                                                                                                                                                                                                                                                                      | MATERIALS | RESOURCES |
| <ul style="list-style-type: none"><li>• D battery</li><li>• Lightbulb</li><li>• Hookup wire</li></ul>                                                                                                                                                                                                                      |           |           |

| GOALS                                                                                                                     | MATERIALS | RESOURCES |
|---------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| <ul style="list-style-type: none"> <li>• <b><u>Activity 3.1.1 Illumination Creation (Downloadable PDF)</u></b></li> </ul> |           |           |

American inventor, Thomas Alva Edison is considered to be one of the most influential inventors in history. Over his lifetime, Edison was awarded 1,093 United States patents that focused on advancements in electricity, recorded sound, telegraphy, batteries, mining, cement, and motion pictures.

Several of Edison's inventions changed the world for the better—including most notably, the incandescent lightbulb. The artificial light the lightbulb generated was used to extend the workday, create night culture, and provide safe home environments for families in the evening. His invention also led to energy developments, from power plants to home appliances and electricity transmission lines.



Although Edison is primarily credited for the invention of the incandescent lightbulb, British inventor, Humphry Davy, first created light with the arc lamp approximately 40 years earlier. Edison studied Davy's designs and made improvements until the incandescent lightbulb was cost-effective and practical for commercial and personal use.

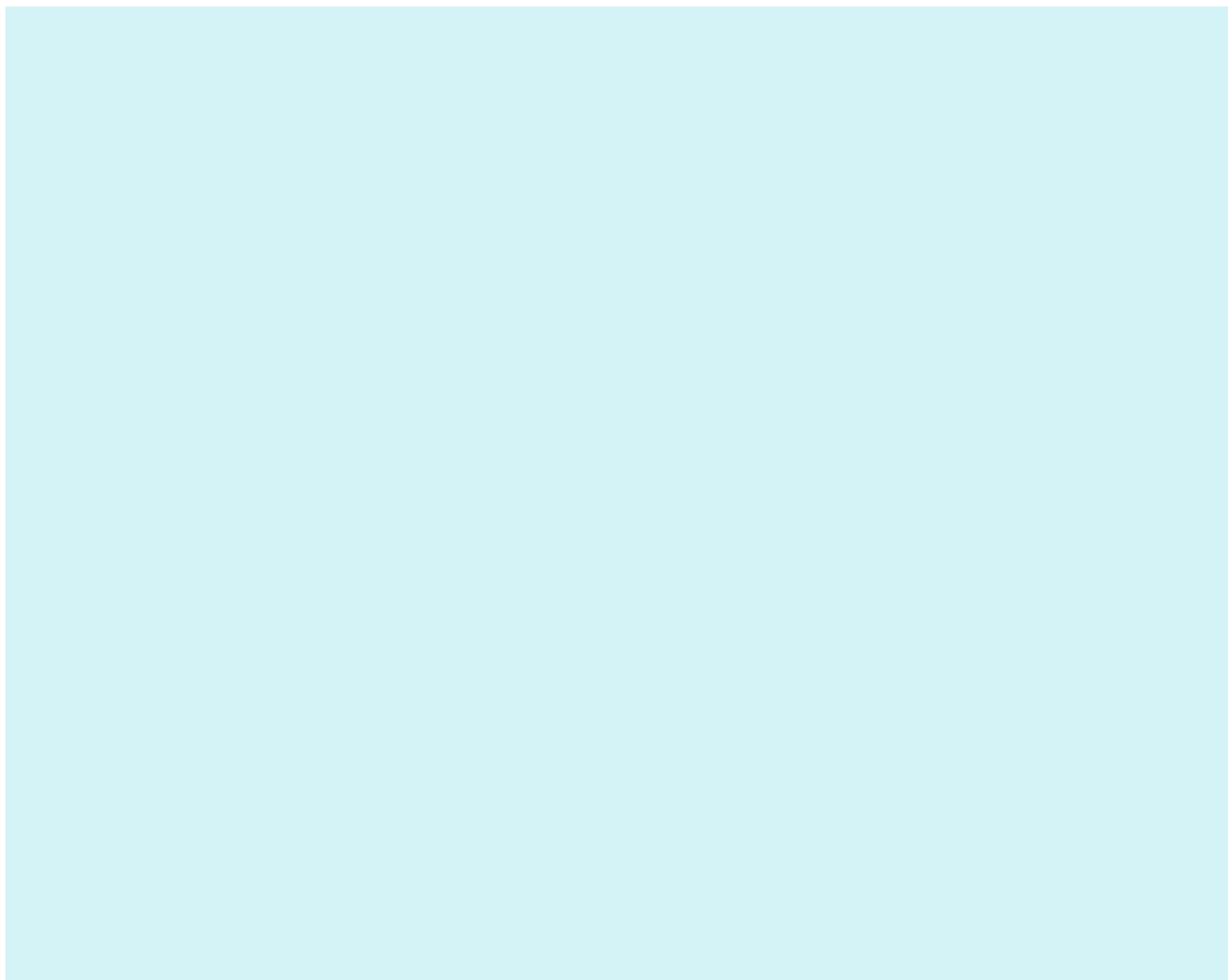
In this activity, you'll explore a simple electrical circuit using an incandescent bulb similar to the design Edison patented more than 100 years ago.

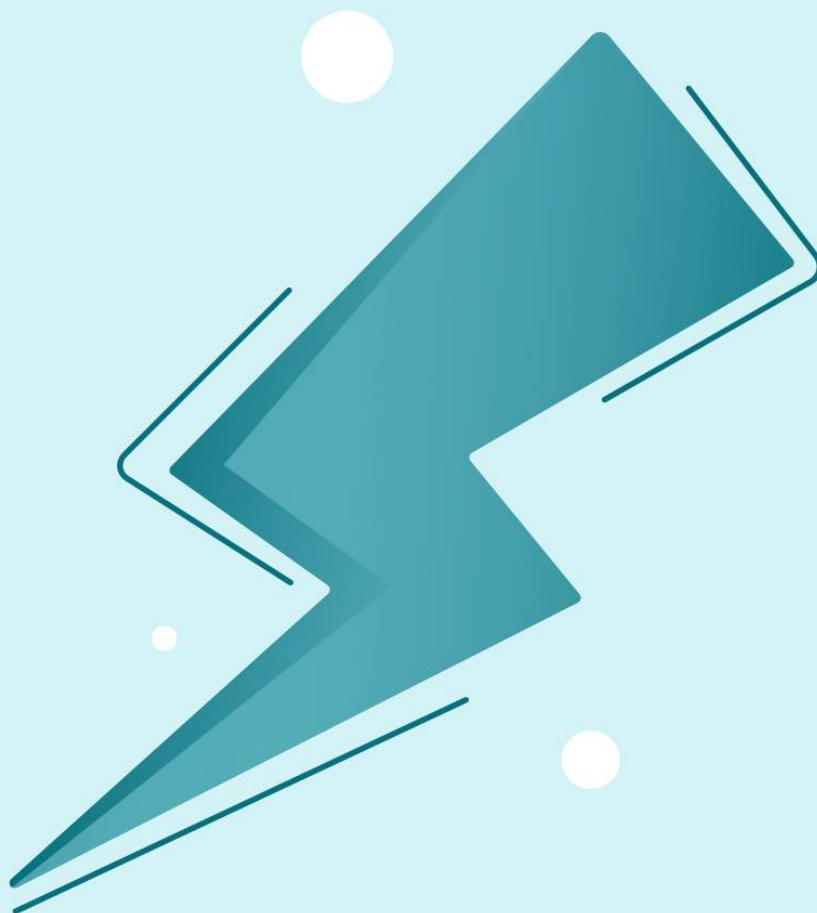
# Building Circuits

---

1

With a partner, review the challenge materials and instructions, make a plan, and get started!





## INSTANT CHALLENGE

**Your Goal:** Using only one battery, one lightbulb, and one wire, find four different configurations that cause the lightbulb to light up.

2

Draw all circuit attempts—both successful and unsuccessful, in your PLTW Engineering Notebook. Your drawings must:

- Show the difference between the two ends of the battery.
- Show where the lightbulb is touching the wire and the battery.

Figure 1 shows simple ways to draw the battery, lightbulb, and wire in your notebook.



Figure 1. Example Sketches

## Reflection

As you and your partner experimented with different electrical configurations, you likely experienced both failure and success.

3

Generate ideas that explain why some configurations work while others do not. As you reflect on your work, record your first thoughts and impressions.

## **Gavin Bence** | *Electrical Engineer*

Electricity is a component in many products and design systems, so an engineer needs to have a solid foundation of electrical knowledge.

Gavin Bence was a Project Lead the Way student in high school, taking Introduction to Engineering Design, Principles of Engineering, and Digital Electronics. These courses led him to major in electrical engineering and minor in entertainment engineering in college.

Gavin now manages a cross-functional team of mechanical, electrical, and materials engineers that produce industrial warning labels for industrial machinery, tractors, and photovoltaic systems.





## Unit 3 Problem Introduction

---

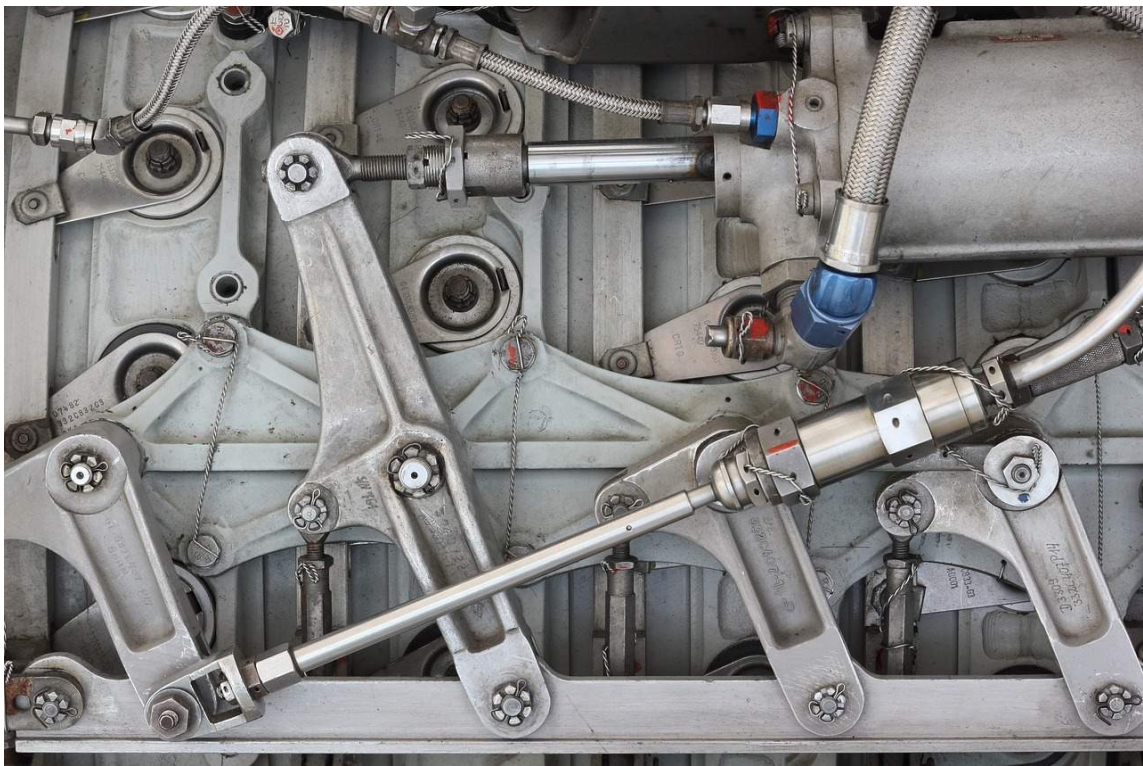
### **Integration of Circuits, Fluid Power, and Kinematics**

Physics helps us understand the rules that govern the universe, covering everything from subatomic particles to the vastness of space. Successful engineers are able to apply physics principles to solve problems, design innovative technologies, and create systems that control our built environment. In this unit, you'll be asked to think like an engineer, answering questions like:

- How do electrical circuits function, and what fundamental principles govern their behavior? How can this understanding be leveraged to analyze and design circuits for specific applications?
- How can the principles of fluid mechanics be used to design and efficiently control hydraulic and pneumatic systems? What considerations are essential for optimizing fluid power systems in various engineering contexts?

- How can the laws that explain bodies in motion be applied to analyze and design mechanical systems, considering factors such as velocity, acceleration, and displacement, to optimize performance and efficiency?

Reflect on these questions as you progress through Unit 3. Each activity and project prepares you for solving the final unit problem of developing a device to automate a popular game.



# Conclusion

---

## Question 1

Based on your experience in this challenge, what do you think is the difference between an open and a closed circuit?

## Question 2

Look at some of the drawings your classmates created. Were you able to clearly understand their design intent? Is there a way to standardize drawings so that anyone, anywhere can read and understand them?